Things I Don't Understand About Quantum Error Correction

A naive audit of the constructs — the confusions of a self-taught researcher, treated as data

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Poster Session: Wed 26 Aug, 16:30–18:00 · Rowan Brad Quni-Gudzinas · Independent Researcher

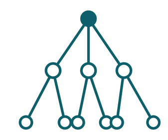
Panel 1 — Why this poster exists

I am an independent researcher, not a member of any quantum-error-correction group. I came to QEC from the outside — from ultrametric geometry and the energy cost of computation. On the way in, I collected **eight questions I cannot answer myself**.

This poster is my notebook made public: eight questions stated as questions. I am not claiming the field is wrong — I am asking where the pieces were checked against each other, and I would like the room's help with the answers.

The invitation: answer any of the eight questions and you will have taught me something real. Correct me, and the poster has done its job.

Panel 2 — The Method: questions as probe



I drew the structure I kept reading about — the Bruhat–Tits tree T_2 (p=2, 3-regular branching; the full construction truncates at depth 3, giving 22 vertices and 21 edges). Drawing a structure is when its assumptions become visible.

I ask the questions I'm not supposed to need to ask. The naive question is the **external structural probe**: it finds the locales and the seams. Where imports contradict, understanding breaks — **that break is data, not deficit.**

Method lineage: IAPS §6.2 audit-the-map; the Universal Ignorance Audit (15 questions, 10.5281/zenodo.21901984); "Knowing What We Do Not Know" (10.5281/zenodo.21901983).

Panel 3 — Things I don't understand about QEC

1 · Why the discreteness?
QEC protects a discrete abstraction (gates, qubits, circuits). Is the discreteness physical — or imported from the circuit model? If the abstraction is a choice, fault tolerance is the tax on the choice.

2 · What is a logical qubit, physically?
Hardware is bosonic oscillators with leakage levels; codes assume qubits. The map is 2D; the machine is continuous. Where does the map end?

3 · What does a threshold actually guarantee?
The proofs assume stochastic local noise. Coherent and correlated errors are the seams nobody prices. What is the guarantee outside the model?

4 · Where is the quantum in the loop?
Syndrome → *classical* decoder → feedback. QEC is quantum memory + classical control. Why is it called quantum error correction?

5 · Why isn't the decoder part of the code?
A classical algorithm decides a quantum code's performance. Coding theory and classical computation — two silos, never audited for compatibility.

Panel 3 (cont.) — The questions (2 of 2)

6 · Why 2D geometry?
Why a surface code and not a tree? Noise geometry is assumed continuous and Euclidean. What if error propagation is hierarchical — ultrametric (Bruhat–Tits)? The staircase I could not find examined anywhere.

7 · Why count T gates, not joules?
The field prices the algebraic resource (T-count, magic states). The energy of the ancilla factory, verification retries, the decoder, and the feedback loop is nowhere priced. What does a correct answer cost in energy?

8 · Why is the cat the unit of protection?
Shor states are cat states — exponentially fragile to dephasing; verified ancillas go stale between verification and use. Why build protection from something that fragile?

Each question is a probe: the answer either closes the seam (the imports were compatible all along) or reveals it (the construct carried more than its proof).

Panel 4 — The seams: where the map ends

Seam	The contradiction
Discrete abstraction	Continuous physics → discrete gates → QEC defends the discreteness. The defense is priced; the choice is not.
Qubit / boson	The model is finite-dimensional; the hardware leaks. Every code assumes what the hardware denies.
Noise model	Stochastic-local assumptions meet coherent, correlated, adversarial reality. Thresholds live inside the model.
Classical control	The decoder is classical. The "quantum" loop runs on a classical brain — the field knows, and doesn't price it.
Cost currency	T-count is countable, so it is counted. Joules are measurable, and not measured. The algebraic unit stands in for the thermodynamic one.
Geometry	Euclidean 2D assumed; ultrametric tree unexamined. The staircase signature I could not find examined anywhere.

What I do understand: the maps are excellent at what they map. The seam is not the diagram — it is the **silence about where the diagram ends**.

Panel 5 — The statement, the caption, and the questions for the room

The one-line statement: "I cannot grasp quantum computing through the practitioners' constructs because the constructs are maps drawn by different cartographers — internally consistent, mutually contradictory. **My confusions are the fault lines.**"

The caption (feigned naivete is a method, not a pose): the naive question is the probe that finds the seam. I am not claiming the field is wrong — I am claiming the imports were never checked against each other. *If the constructs were compatible, the naive questions would answer themselves.*

Questions I will ask this week:

- "Is the discreteness of the circuit model physical or imported?"
- "Has anyone priced the energy of the decoder — per correct solution?"
- "Why a 2D cluster and not a tree?" (Bruhat–Tits resource states)

References (verified 2026-08-18; updated 2026-08-21)
Universal Ignorance Audit — 10.5281/zenodo.21901984 · Knowing What We Do Not Know — 10.5281/zenodo.21901983
Ultrametric Code Spaces — 10.5281/zenodo.21824194 · JPCUB (joules-per-solution) — 10.5281/zenodo.21945415
Tree-code testing roadmap — QNFO/cwi-qec-poster-2026 · poster-v5-treecode

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